LAB 06-24K0025-Syed Hanzala Ali

Worksheet | Query Builder

```sql
-- Q1. a
SELECT first_name, last_name, hire_date
FROM employees
WHERE hire_date > (
    SELECT MAX(hire_date)
    FROM employees
    WHERE department_id = 60
);
```

Script Output | Query Result

SQL | All Rows Fetched: 21 in 0.003 seconds

	FIRST_NAME	LAST_NAME	HIRE_DATE
1	Luis	Popp	07-DEC-07
2	Karen	Colmenares	10-AUG-07
3	Kevin	Mourgos	16-NOV-07
4	Steven	Markle	08-MAR-08
5	Ki	Gee	12-DEC-07
6	Hazel	Philtanker	06-FEB-08
7	Gerald	Cambrault	15-OCT-07
8	Eleni	Zlotkey	29-JAN-08
9	Oliver	Tuvault	23-NOV-07
10	Mattea	Marvins	24-JAN-08
11	David	Lee	23-FEB-08
12	Sundar	Ande	24-MAR-08
13	Amit	Banda	21-APR-08
14	Sundita	Kumar	21-APR-08
15	Kimberely	Grant	24-MAY-07
16	Charles	Johnson	04-JAN-08
17	Martha	Sullivan	21-JUN-07
18	Girard	Geoni	03-FEB-08
19	Randall	Perkins	19-DEC-07
20	Donald	OConnell	21-JUN-07
21	Douglas	Grant	13-JAN-08

Worksheet | Query Builder

```sql
-- Q1. b
SELECT first_name, last_name, salary
FROM employees
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE job_id = 'SA_REP'
);
```

Script Output | Query Result

SQL | All Rows Fetched: 30 in 0.004 seconds

	FIRST_NAME	LAST_NAME	SALARY
1	Steven	King	24000
2	Neena	Kochhar	17000
3	Lex	De Haan	17000
4	Alexander	Hunold	9000
5	Nancy	Greenberg	12008
6	Daniel	Faviet	9000
7	Den	Raphaely	11000
8	John	Russell	14000
9	Karen	Partners	13500
10	Alberto	Errazuriz	12000
11	Gerald	Cambrault	11000
12	Eleni	Zlotkey	10500
13	Peter	Tucker	10000
14	David	Bernstein	9500
15	Peter	Hall	9000
16	Janette	King	10000
17	Patrick	Sully	9500
18	Allan	McEwen	9000
19	Clara	Vishney	10500
20	Danielle	Greene	9500
21	Lisa	Ozer	11500
22	Harrison	Bloom	10000
23	Tayler	Fox	9600
24	Ellen	Abel	11000
25	Alyssa	Hutton	8800
26	Jonathon	Taylor	8600
27	Jack	Livingston	8400
28	Michael	Hartstein	13000
29	Hermann	Baer	10000
30	Shelley	Higgins	12008

Worksheet | Query Builder

```sql
-- Q1. c
SELECT first_name, last_name, department_id
FROM employees
WHERE department_id = (
    SELECT department_id
    FROM employees
    WHERE employee_id = 103
)
AND employee_id != 103;
```

Script Output | Query Result

SQL | All Rows Fetched: 4 in 0.002 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
1	Bruce	Ernst	60
2	David	Austin	60
3	Valli	Pataballa	60
4	Diana	Lorentz	60

Worksheet | Query Builder

```sql
-- Q1. d
SELECT job_title, min_salary
FROM jobs
WHERE min_salary > (
    SELECT AVG(salary)
    FROM employees
);
```

Script Output | Query Result

SQL | All Rows Fetched: 7 in 0.002 seconds

	JOB_TITLE	MIN_SALARY
1	President	20080
2	Administration Vice President	15000
3	Finance Manager	8200
4	Accounting Manager	8200
5	Sales Manager	10000
6	Purchasing Manager	8000
7	Marketing Manager	9000

Worksheet | Query Builder

```sql
-- Q1. e
SELECT first_name, last_name, salary, department_id
FROM employees
WHERE salary IN (
    SELECT MIN(salary)
    FROM employees
    GROUP BY department_id
);
```

Script Output | Query Result

SQL | All Rows Fetched: 26 in 0.017 seconds

	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
1	Neena	Kochhar	17000	90
2	Lex	De Haan	17000	90
3	Bruce	Ernst	6000	60
4	Diana	Lorentz	4200	60
5	Luis	Popp	6900	100
6	Karen	Colmenares	2500	30
7	Shanta	Vollman	6500	50
8	James	Marlow	2500	50
9	TJ	Olson	2100	50
10	Joshua	Patel	2500	50
11	Peter	Vargas	2500	50
12	Peter	Tucker	10000	80
13	Oliver	Tuvault	7000	80
14	Janette	King	10000	80
15	Sarath	Sewall	7000	80
16	Harrison	Bloom	10000	80
17	Sundita	Kumar	6100	80
18	Kimberely	Grant	7000	(null)
19	Martha	Sullivan	2500	50
20	Nandita	Sarchand	4200	50
21	Randall	Perkins	2500	50
22	Jennifer	Whalen	4400	10
23	Pat	Fay	6000	20
24	Susan	Mavris	6500	40
25	Hermann	Baer	10000	70
26	William	Gietz	8300	110

Worksheet | Query Builder

```sql
-- Q2. a
SELECT *
FROM employees
WHERE manager_id = (
    SELECT manager_id
    FROM employees
    WHERE employee_id = 101
);
```

Script Output | Query Result

SQL | All Rows Fetched: 14 in 0.028 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
2	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
3	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
4	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
5	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
6	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
7	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
8	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
9	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
10	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
11	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
12	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
13	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
14	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20

Worksheet | Query Builder

```sql
-- Q2. b
SELECT *
FROM employees e
WHERE (
    SELECT COUNT(DISTINCT manager_id)
    FROM employees
    WHERE department_id = e.department_id
) > 1;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.007 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
2	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
3	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
4	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
5	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
6	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
7	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
8	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
9	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
10	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
11	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
12	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
13	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3100	(null)	114	30
14	116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	2900	(null)	114	30
15	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800	(null)	114	30
16	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600	(null)	114	30
17	119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2500	(null)	114	30
18	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
19	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
20	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
21	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
22	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
23	125	Julia	Nayer	JNAYER	650.124.1214	16-JUL-05	ST_CLERK	3200	(null)	120	50
24	126	Irene	Mikkilineni	IMIKKILI	650.124.1224	28-SEP-06	ST_CLERK	2700	(null)	120	50
25	127	James	Landry	JLANDRY	650.124.1334	14-JAN-07	ST_CLERK	2400	(null)	120	50
26	128	Steven	Markle	SMARKLE	650.124.1434	08-MAR-08	ST_CLERK	2200	(null)	120	50
27	129	Laura	Bissot	LBISSOT	650.124.5234	20-AUG-05	ST_CLERK	3300	(null)	121	50
28	130	Mozhe	Atkinson	MATKINSO	650.124.6234	30-OCT-05	ST_CLERK	2800	(null)	121	50
29	131	James	Marlow	JAMRLOW	650.124.7234	16-FEB-05	ST_CLERK	2500	(null)	121	50
30	132	TJ	Olson	TJOLSON	650.124.8234	10-APR-07	ST_CLERK	2100	(null)	121	50
31	133	Jason	Mallin	JMALLIN	650.127.1934	14-JUN-04	ST_CLERK	3300	(null)	122	50
32	134	Michael	Rogers	MROGERS	650.127.1834	26-AUG-06	ST_CLERK	2900	(null)	122	50
33	135	Ki	Gee	KGEE	650.127.1734	12-DEC-07	ST_CLERK	2400	(null)	122	50

Worksheet | Query Builder

```sql
-- Q2. c
SELECT department_id
FROM employees
GROUP BY department_id
HAVING AVG(salary) > (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = 90
);
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.002 seconds

DEPARTM...

```sql
-- Q2. d
SELECT *
FROM employees
WHERE salary > (
    SELECT MEDIAN(salary)
    FROM employees
);
```

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
6	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
7	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
8	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
9	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
10	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
11	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
12	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
13	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
14	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
15	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
16	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
17	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
18	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
19	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
20	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
21	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
22	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
23	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
24	153	Christopher	Olsen	COLSEN	011.44.1344.498718	30-MAR-06	SA_REP	8000	0.2	145	80

Worksheet | Query Builder

```sql
-- Q2. e
SELECT *
FROM employees
WHERE commission_pct > (
    SELECT AVG(commission_pct)
    FROM employees
);
```

Script Output | Query Result

All Rows Fetched: 17 in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
2	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
3	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
4	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
5	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
6	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
7	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
8	156	Janette	King	JKING	011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146	80
9	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
10	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
11	159	Lindsey	Smith	LSMITH	011.44.1345.729268	10-MAR-05	SA_REP	8000	0.3	146	80
12	160	Louise	Doran	LDORAN	011.44.1345.629268	15-DEC-05	SA_REP	7500	0.3	146	80
13	161	Sarath	Sewall	SSEWALL	011.44.1345.529268	03-NOV-06	SA_REP	7000	0.25	146	80
14	162	Clara	Vishney	CVISHNEY	011.44.1346.129268	11-NOV-05	SA_REP	10500	0.25	147	80
15	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148	80
16	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80
17	175	Alyssa	Hutton	AHUTTON	011.44.1644.429266	19-MAR-05	SA_REP	8800	0.25	149	80

Worksheet | Query Builder

```sql
SELECT *
FROM employees
WHERE department_id IN (
    SELECT department_id
    FROM departments
    WHERE location_id IN (
        SELECT location_id
        FROM departments
        WHERE department_name = 'Marketing'
    )
);
```

Script Output | Query Result

All Rows Fetched: 2 in 0.158 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
2	202	Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6000	(null)	201	20

Worksheet | Query Builder

```sql
-- Q2. g
SELECT *
FROM employees
WHERE salary NOT IN (
    SELECT salary
    FROM employees
    WHERE department_id = 50
    AND salary IS NOT NULL
);
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
7	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
8	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
9	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
10	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
11	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
12	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
13	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
14	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
15	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
16	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
17	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
18	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
19	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
20	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
21	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
22	154	Nanette	Cambrault	NCAMBRAU	011.44.1344.987668	09-DEC-06	SA_REP	7500	0.2	145	80
23	155	Oliver	Tuvault	OTUVAULT	011.44.1344.486508	23-NOV-07	SA_REP	7000	0.15	145	80
24	156	Janette	King	JKING	011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146	80
25	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
26	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
27	160	Louise	Doran	LDORAN	011.44.1345.629268	15-DEC-05	SA_REP	7500	0.3	146	80
28	161	Sarath	Sewall	SSEWALL	011.44.1345.529268	03-NOV-06	SA_REP	7000	0.25	146	80
29	162	Clara	Vishney	CVISHNEY	011.44.1346.129268	11-NOV-05	SA_REP	10500	0.25	147	80
30	163	Danielle	Greene	DGREENE	011.44.1346.229268	19-MAR-07	SA_REP	9500	0.15	147	80
31	164	Mattea	Marvins	MMARVINS	011.44.1346.329268	24-JAN-08	SA_REP	7200	0.1	147	80
32	165	David	Lee	DLEE	011.44.1346.529268	23-FEB-08	SA_REP	6800	0.1	147	80
33	166	Sundar	Ande	SANDE	011.44.1346.629268	24-MAR-08	SA_REP	6400	0.1	147	80

Worksheet | Query Builder

```sql
-- Q2. h
SELECT *
FROM employees
WHERE salary BETWEEN (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = 10
) AND (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = 20
);
```

Script Output | Query Result

All Rows Fetched: 41 in 0.021 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
2	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
3	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
4	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
5	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
6	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
7	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
8	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
9	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
10	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
11	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
12	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
13	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
14	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
15	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
16	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
17	153	Christopher	Olsen	COLSEN	011.44.1344.498718	30-MAR-06	SA_REP	8000	0.2	145	80
18	154	Nanette	Cambrault	NCAMBRAU	011.44.1344.987668	09-DEC-06	SA_REP	7500	0.2	145	80
19	155	Oliver	Tuvault	OTUVAULT	011.44.1344.486508	23-NOV-07	SA_REP	7000	0.15	145	80
20	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
21	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
22	159	Lindsey	Smith	LSMITH	011.44.1345.729268	10-MAR-05	SA_REP	8000	0.3	146	80
23	160	Louise	Doran	LDORAN	011.44.1345.629268	15-DEC-05	SA_REP	7500	0.3	146	80
24	161	Sarath	Sewall	SSEWALL	011.44.1345.529268	03-NOV-06	SA_REP	7000	0.25	146	80
25	163	Danielle	Greene	DGREENE	011.44.1346.229268	19-MAR-07	SA_REP	9500	0.15	147	80
26	164	Mattea	Marvins	MMARVINS	011.44.1346.329268	24-JAN-08	SA_REP	7200	0.1	147	80

Worksheet | Query Builder

```sql
-- Q2. i
SELECT *
FROM employees
WHERE employee_id IN (
    SELECT employee_id
    FROM job_history
);
```

Script Output | Query Result

All Rows Fetched: 7 in 0.161 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
2	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
3	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
4	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
5	176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	24-MAR-06	SA_REP	8600	0.2	149	80
6	200	Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-03	AD_ASST	4400	(null)	101	10
7	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20

Worksheet | Query Builder

```sql
-- Q3
SELECT *
FROM employees e
WHERE department_id IN (
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING AVG(salary) > (SELECT AVG(salary) FROM employees)
)
AND salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = e.department_id
)
AND (EXTRACT(YEAR FROM SYSDATE) - EXTRACT(YEAR FROM hire_date)) > 10
AND NOT EXISTS (
    SELECT 1
    FROM job_history jh
    WHERE jh.employee_id = e.employee_id
)
AND NOT EXISTS (
    SELECT 1
    FROM employees m
    WHERE m.manager_id = e.employee_id
);
```

Script Output | Query Result

SQL | All Rows Fetched: 13 in 0.036 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148	80
2	162	Clara	Vishney	CVISHNEY	011.44.1346.129268	11-NOV-05	SA_REP	10500	0.25	147	80
3	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
4	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
5	156	Janette	King	JKING	011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146	80
6	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
7	170	Tayler	Fox	TFOX	011.44.1343.729268	24-JAN-06	SA_REP	9600	0.2	148	80
8	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
9	163	Danielle	Greene	DGREENE	011.44.1346.229268	19-MAR-07	SA_REP	9500	0.15	147	80
10	169	Harrison	Bloom	HBLOOM	011.44.1343.829268	23-MAR-06	SA_REP	10000	0.2	148	80
11	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
12	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
13	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80

Welcome Page | hr

Worksheet | Query Builder

```sql
-- Q4
SELECT department_id
FROM departments d
WHERE EXISTS (
    SELECT 1
    FROM employees
    WHERE department_id = d.department_id
    AND salary > (SELECT AVG(salary) FROM employees)
)
AND EXISTS (
    SELECT 1
    FROM employees
    WHERE department_id = d.department_id
    AND salary < (SELECT AVG(salary) FROM employees)
);
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.015 seconds

	DEPARTMENT_ID
1	60
2	30
3	50
4	80
5	20

Worksheet | Query Builder

```sql
-- Q5
SELECT *
FROM employees e
WHERE department_id IN (
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING SUM(salary) > (SELECT MAX(salary) FROM employees)
)
AND salary < (
    SELECT MAX(salary)
    FROM employees
    WHERE department_id = e.department_id
)
AND salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = e.department_id
)
AND NOT EXISTS (
    SELECT 1
    FROM employees m
    WHERE m.manager_id = e.employee_id
)
AND salary > ALL (
    SELECT salary
    FROM employees
    WHERE department_id = 30
    AND salary IS NOT NULL
)
AND department_id IN (
    SELECT department_id
    FROM departments
    WHERE location_id IN (
        SELECT location_id
        FROM departments
        GROUP BY location_id
        HAVING COUNT(department_id) > 1
    )
);
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.055 seconds

EMPLOYE...	FIRST_NA...	LAST_NAME	EMAIL	PHONE_N...	HIRE_DATE	JOB_ID	SALARY	COMMISS...	MANAGER...	DEPARTM...

Worksheet | Query Builder

```sql
-- Q6
SELECT *
FROM employees e
WHERE department_id IN (
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING (MAX(salary) - MIN(salary)) > (SELECT AVG(salary) FROM employees)
)
AND salary > (
    SELECT MEDIAN(salary)
    FROM employees
    WHERE department_id = e.department_id
)
AND salary < (
    SELECT MAX(salary)
    FROM employees
    WHERE department_id = e.department_id
)
AND NOT EXISTS (
    SELECT 1
    FROM job_history jh
    WHERE jh.employee_id = e.employee_id
)
AND department_id IN (
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING MIN(salary) > 4000
);
```

Script Output | Query Result

SQL | All Rows Fetched: 16 in 0.043 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
2	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
3	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
4	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
5	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
6	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
7	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
8	156	Janette	King	JKING	011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146	80
9	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
10	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
11	162	Clara	Vishney	CVISHNEY	011.44.1346.129268	11-NOV-05	SA_REP	10500	0.25	147	80
12	163	Danielle	Greene	DGREENE	011.44.1346.229268	19-MAR-07	SA_REP	9500	0.15	147	80
13	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148	80
14	169	Harrison	Bloom	HBLOOM	011.44.1343.829268	23-MAR-06	SA_REP	10000	0.2	148	80
15	170	Tayler	Fox	TFOX	011.44.1343.729268	24-JAN-06	SA_REP	9600	0.2	148	80
16	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80

Worksheet | Query Builder

```sql
-- Q7
SELECT *
FROM employees e
WHERE department_id IN (
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING AVG(salary) > (SELECT AVG(salary) FROM employees)
)
AND salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = e.department_id
)
AND salary < (
    SELECT MAX(salary)
    FROM employees
)
AND salary > ALL (
    SELECT salary
    FROM employees
    WHERE department_id = 20
    AND salary IS NOT NULL
)
AND hire_date < (
    SELECT m.hire_date
    FROM employees m
    WHERE m.employee_id = e.manager_id
)
AND EXISTS (
    SELECT 1
    FROM employees
    WHERE department_id = e.department_id
    AND EXTRACT(YEAR FROM hire_date) > 2020
)
AND NOT EXISTS (
    SELECT 1
    FROM employees m
    WHERE m.manager_id = e.employee_id
)
AND NOT EXISTS (
    SELECT 1
    FROM job_history jh
    WHERE jh.employee_id = e.employee_id
);
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.039 seconds

EMPLOYE...	FIRST_NA...	LAST_NAME	EMAIL	PHONE_N...	HIRE_DATE	JOB_ID	SALARY	COMMISS...	MANAGER...	DEPARTM...